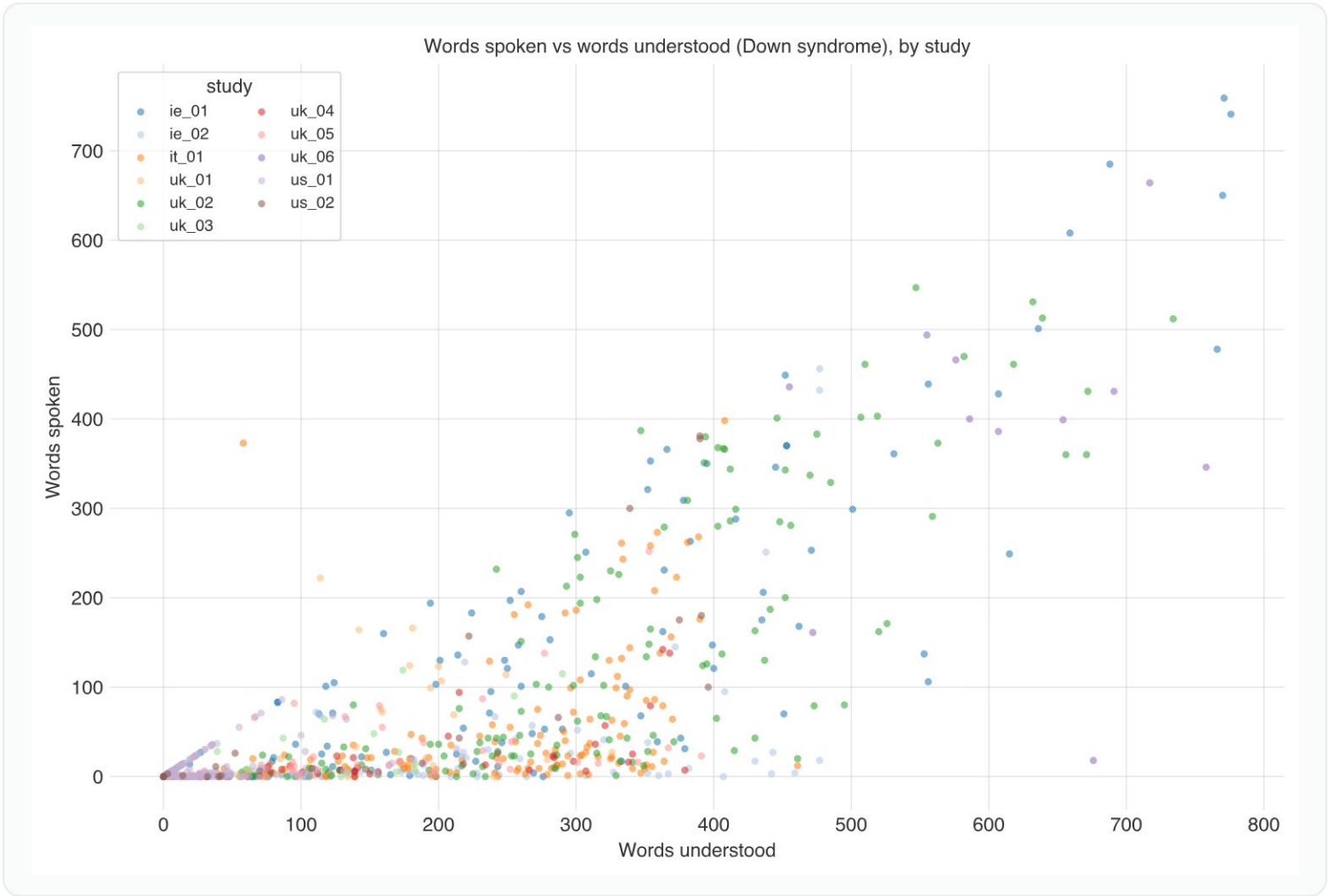


How children with Down syndrome build vocabulary

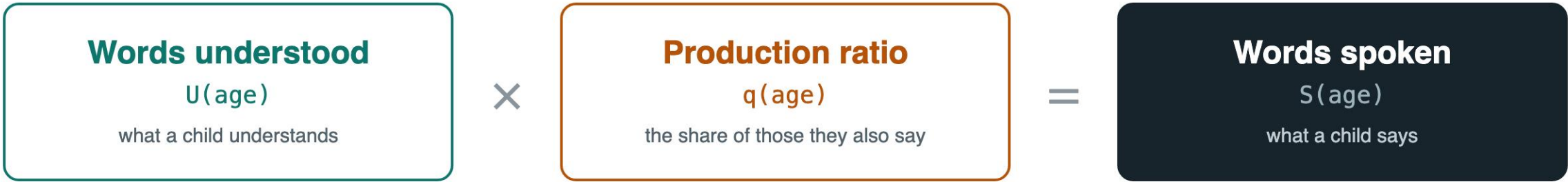
A pooled Bayesian study of the words children understand, say and sign, across eleven international datasets.



Words spoken against words understood, one point per assessment, coloured by study. Almost every point sits below the diagonal: children understand more than they say.

How the vocabulary model works

The modelling idea at the centre of the study: spoken vocabulary is built from what a child understands and the share of it they say.

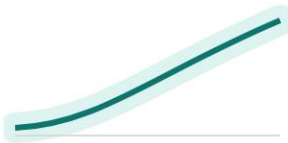


Modelling the two together means spoken vocabulary can **never exceed** what a child understands, and the comprehension–production gap falls out directly as the production ratio **q**.

THE DATA

Parent-report vocabulary checklists, **pooled across 11 international studies** onto a common 800-word scale: **588 children**, ages 8 months to about 9½ years, roughly 1,000 observations.

THE ESTIMATE

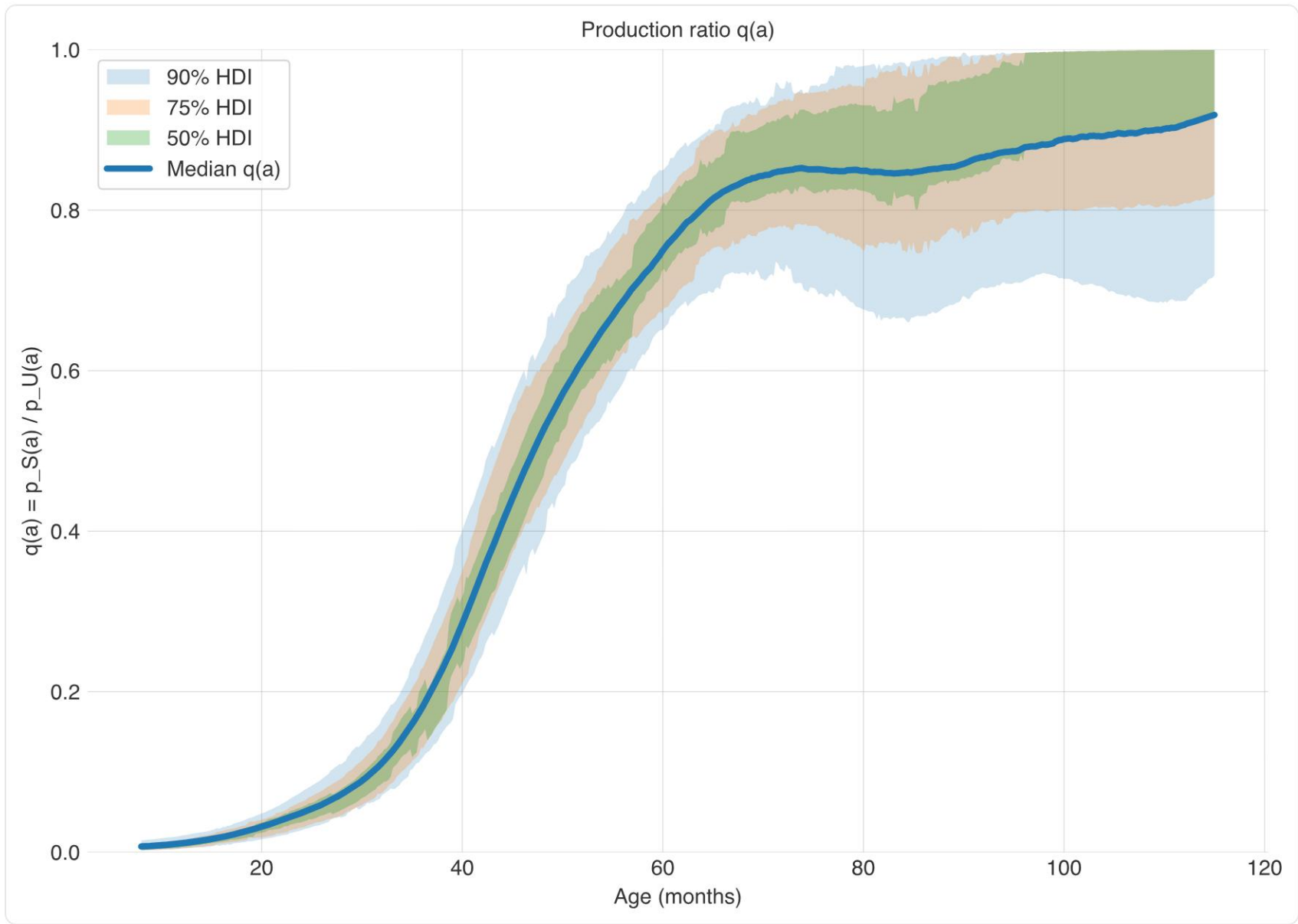


Each term is a **smooth curve over age** with full posterior uncertainty, so every result is a credible range, not a single number.

- Hierarchical Beta–Binomial
- Gaussian–process (HSGP) trends
- study & child random effects
- PyMC / NUTS
- PSIS–L00 comparison

Understanding runs years ahead of speaking

The share of understood words a child with Down syndrome also says, estimated across age from pooled international data.



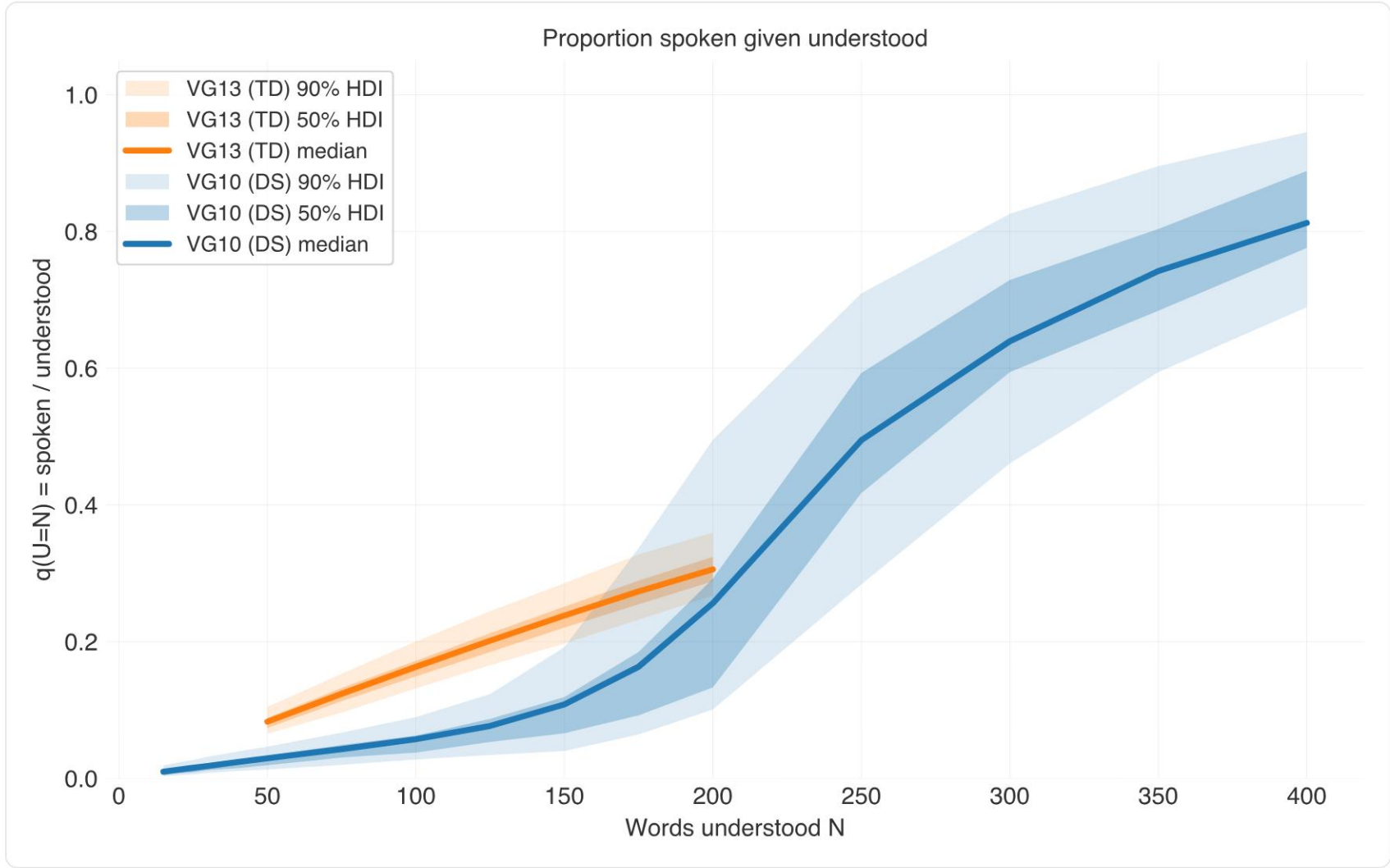
Each child's **production ratio** is the fraction of the words they understand that they also say. It climbs slowly — comprehension leads production for years.

~1% **~52%** **~86%**
at 1 year at 4 years by 7½ years

Wider than in typically-developing children at low vocabulary levels, narrowing as vocabulary grows — evidence the gap is a genuine stretch, not just a delay.

Children with Down syndrome say less of what they understand

Comparing the production ratio at matched comprehension, against typically-developing children.



At the same number of words understood, **typically-developing children** say a larger share of them than **children with Down syndrome** — most of all when vocabulary is small.

+13 pts

peak gap in share spoken

50–150

words understood, where the gap is credible

The gap narrows as vocabulary grows, converging toward the typically-developing pattern by about 200 understood words. It is a stretch that eases, not a fixed deficit.